

Algorithms and Responsibility (#133980)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
2023/05/30 16:27 (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether:

RQ 1. punishment for a decision can be avoided if it is delegated to an algorithm (conditional on performance)

RQ 2. a punishment motive enters the delegation decision.

3) Describe the key dependent variable(s) specifying how they will be measured.

The experiment features a delegator and an evaluator. The delegator has to make a decision that is payoff-relevant only to the evaluator. The delegator may decide herself or delegate the decision to an algorithm, which is calibrated to match the delegators performance. Depending on the treatment, the evaluator may punish the delegator conditional on the outcome of the decision and the delegation decision (strategy method). For RQ 1, we will look at evaluator punishment behavior and compare the amount and frequency of punishment between a delegated and non-delegated decision conditional on the outcome.

For RQ 2, we will look at the delegation frequency between the treatments (punishment possible vs punishment not possible).

4) How many and which conditions will participants be assigned to?

There are two conditions (punishment possible/no punishment possible) and two roles in each condition (delegator/evaluator).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

I will run parametric (t-test) and non-parametric (MWU, Kolmogorov-Smirnov) tests between punishment (amount and frequency) for delegated and non-delegated outcomes, conditional on the outcome. Similarly, I will run linear regressions with punishment behavior as the dependent variable and explore whether expected performance or any socio-demographic characteristics are predictive of differential punishment.

For delegators, I will again, run parametric (t-test) tests on delegation choices between treatments. Moreover, I will employ Probit regressions with the dependent variable being the delegation decision and a set of explanatory variables including treatment, beliefs about performance and socio-demographic characteristics.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

I will exclude people that access the experiment via mobile device. Moreover, I may exclude participants that failed the attention checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

I will recruit subjects via Prolific. I plan to recruit 320 participants, 160 per treatment, 80 per role per treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

I already ran a pilot with 25 observations in the punishment treatment. I plan to include the data in the analysis.

Secondary analysis will include an investigation on how performance and beliefs about performance impact results. Moreover, I will look at the delegator's beliefs about expected punishment and their consistency with the delegation decision.

At this point, I already want to state that I may run another treatment in which the delegator may either delegate to another participant or an algorithm.